

Paper 12.75 - CA Prediction Surface as Next-State Precondition

CQE-paper-12.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-12.75.md

Paper 12.75 - CA Prediction Surface as Next-State Precondition

Purpose

Paper 12.75 explains how the CA prediction surface becomes a precondition for Paper 13 and later applied prediction papers.

Exported State

Paper 12 exports:

```
Rule30 local readout
T_EMISSION formula
Rule90 correction decomposition
64 silent-boundary ECA count
layer receipt requirement
open cold-start extractor
empirical spectral layer boundary
```

Use in Paper 13

Paper 13 may use the CA correction field as a quark-face or color transport input only if it preserves the layer receipt and correction status.

The allowed transition is:

```
CA prediction surface receipt -> quark-face transport input
```

The disallowed transition is:

```
empirical spectral guess -> closed particle transport theorem
```

Use in Later Papers

Later papers may use Paper 12 for:

- deterministic local readout,
- correction-field transport,
- CA candidate comparison,
- empirical prediction surfaces,
- product or tool prediction layers.

Every use must state whether it imports the closed local layer or an open prediction layer.

Precondition Rule

Before a later paper uses Paper 12, it should be able to answer:

```
which CA rule is active?
which layer produced the result?
what is the local state or query?
what is the cost class?
what is the defect status?
which receipt proves it?
```

Conclusion

Paper 12.75 turns the CA prediction surface into portable state. It lets later papers use local readout and correction fields while keeping cold-start and spectral prediction claims attached to their own future proofs.